

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Supplementary Examination Winter – 2024			
Course: B. Tech	Branch : Electronics & Computer Engineering		Semester :VI
Subject Code & Name:	BTECPE603D	Software Testing	
Max Marks: 60	Date:23/12/2024		Duration: 3 Hr.

Instructions to the Students:

- Each question carries 12 marks.
- Question No. 1 will be compulsory and include objective-type questions.
- Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
- The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
- Use of non-programmable scientific calculators is allowed.
- Assume suitable data wherever necessary and mention it clearly.

Q.1	This Question is Compulsory , Select Appropriate answers from multiple choice	(Level/CO)	Mark 12
1	Which of the following is NOT a phase in the Software Development Life Cycle (SDLC)? a) Requirement Analysis b) Testing c) Debugging d) Deployment	Remembering /CO1	1
2	What does the "V" in V-Model stand for? a) Verification and Validation b) Value-based Testing c) Versioning d) Virtual Testing	Understanding/CO1	1
3	Which of the following is a part of Quality Assurance (QA)? a) Fixing defects b) Validating user requirements c) Ensuring proper process adherence d) Identifying code errors	Understanding / CO1	1
4	What is another term for White Box Testing? a) Behavioral Testing b) Glass Box Testing c) Functional Testing d) Black Box Testing	Remembering /CO2	1
5	Which of the following techniques is NOT used in Black Box Testing? a) Boundary Value Analysis b) Equivalence Partitioning c) Statement Coverage d) Decision Table Testing	Understanding/CO2	1
6	The primary focus of White Box Testing is on: a) User interface b) Internal code structure	Understanding / CO2	1

	c) Software requirements d) End-user behavior		
7	Which integration testing approach begins testing from the top-level modules? a) Bottom-Up Integration b) System Integration c) Top-Down Integration d) Sandwich Integration	Understanding/CO3	1
8	Defect Bash is commonly conducted during: a) Unit Testing b) System Testing c) Integration Testing d) Acceptance Testing	Remembering /CO3	1
9	Scenario testing focuses on: a) Isolated modules b) Real-world user scenarios c) Boundary conditions d) Unit-level code execution	Applying /CO3	1
10	Non-functional testing evaluates: a) Scalability and performance b) Business requirements c) Functional correctness d) User interface design	Understanding/ CO4	1
11	Acceptance testing is primarily conducted by: a) Test engineers b) Developers c) End users or clients d) Project managers	Remembering /CO4	1
12	What is the main purpose of Regression Testing? a) Identifying new defects in a system b) Ensuring changes do not break existing functionality c) Testing application performance d) Verifying localization features	Understanding /CO4	1
Q2.	Solve the following questions		12
	A) Explain the differences between Verification and Validation in the context of software development. Provide relevant examples for each process.	Understanding / CO1	6
	B) Discuss the various Software Development Life Cycle (SDLC) models. Highlight their suitability and key differences.	Analyzing / CO1	6
Q3.	Solve the following.		12
	A) Explain the key differences between White Box Testing (WBT) and Black Box Testing (BBT). Provide examples of when each testing method is appropriate.	Understanding / CO2	6
	B) Describe the process of Boundary Value Analysis (BVA) in Black Box Testing. Provide an example to illustrate the concept.	Applying / CO2	6
Q4	Solve Any Two of the following.		12

	A) Explain the different types of integration testing approaches with suitable examples.	Understanding / CO3	6
	B) What is Scenario Testing, and how does it differ from other testing methods? Illustrate with an example.	Applying/CO3	6
	C) Discuss the importance of Defect Bash in Integration Testing. Highlight its process and advantages.	Analyzing / CO3	6
Q5	Solve Any Two of the following.		12
	A) Differentiate between Functional and Non-Functional Testing with examples. Explain why both are essential in system testing.	Understanding / CO4	6
	B) What is Acceptance Testing? Explain its types and significance in the software development process.	Applying / CO4	6
	C) Discuss the role of Non-Functional System Testing in evaluating system performance. Provide examples of techniques used.	Analyzing / CO4	6
Q6	Solve Any Two of the following.		12
	A) Explain the methodology of Performance Testing. Why is it important in software development?	Applying/CO5	6
	B) Define Regression Testing. Discuss its types and explain the process involved.	Understanding / CO5	6
	C) What is Adhoc Testing? Describe its types with examples. How does it contribute to overall software quality?	Analyzing / CO5	6

*** All the Best***